# Akoizo Original Practice - 2027 Rotation Form 2

### Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

**Name: _____ Date: _____**

**1. (2 points) A station model pressure code of 984 means:**

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. 1008.4 mb

B. 998.4 mb

C. 1098.4 mb

D. 984 mb

Answer: _____

**2. (2 points) Which air mass is warm and humid?**

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Continental tropical

B. Continental arctic

C. Maritime tropical

D. Continental polar

Answer: _____

**3. (2 points) Damage spreading outward in a starburst pattern indicates:**

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. A tornado

B. Hail

C. A dryline

D. A downburst

Answer: _____

**4. (2 points) Sleet forms when:**

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Melted snow refreezes in a deep cold layer near the ground

B. Snow never melts

C. Rain freezes on contact

D. Hail falls from a supercell

Answer: _____

**5. (2 points) In a METAR, “31/22” means:**

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. Temperature 31 °C, dew point 22 °C

B. Wind 310° at 22 knots

C. Visibility 31 miles

D. Pressure 31.22 inches

Answer: _____

**6. (2 points) A dryline separates air masses that differ mainly in:**

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Wind speed only

B. Moisture

C. Temperature

D. Pressure only

Answer: _____

**7. (2 points) The EF scale estimates wind speed from:**

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. Radar reflectivity

B. Funnel width

C. Observed damage to indicators

D. Direct anemometer readings in the tornado

Answer: _____

**8. (2 points) A tornado warning means:**

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Conditions may become favorable later

B. The storm has ended

C. Only hail is expected

D. A tornado is occurring or imminent; take shelter now

Answer: _____

**9. (2 points) A long hodograph that curves clockwise near the ground indicates:**

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. No wind

B. Stable air

C. Calm conditions favoring fog

D. Strong shear favoring supercells

Answer: _____

**10. (2 points) Why do hurricanes weaken after landfall?**

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. They lose their warm-water energy source

B. Land air is too moist

C. The jet stream stops them

D. Coriolis force becomes zero

Answer: _____

**11. (2 points) Interpret the adjacent opposite radial velocities cautiously.**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

**12. (2 points) What additional interpretation is supported by low correlation coefficient collocated with the velocity couplet?**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

**13. (2 points) Calculate the surface temperature-dew-point spread.**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

**14. (2 points) Identify one reason that 65 dBZ reflectivity alone does not prove a tornado.**

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 27 C
Dew point | 22 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

**15. (2 points) A storm moves 30 km east in 30 minutes. Find mean speed.**

Storm-track table:
Time | Eastward distance
0 min | 0 km
30 min | 30 km

Answer: _____
_____
_____

**16. (2 points) A rain gauge records 40 mm in 0.5 hours. Calculate mean rain rate.**

Gauge table:
Elapsed time | Accumulated rain
0 h | 0 mm
0.5 h | 40 mm

Answer: _____
_____
_____

**17. (2 points) A satellite IR time series shows cloud-top brightness temperature falling from 240 K to 220 K. Give a physically plausible interpretation and a limitation.**

IR table:
Frame 1 | 240 K
Frame 2 | 220 K

Answer: _____
_____
_____

**18. (2 points) Station wind shifts form a coherent boundary while pressure rises behind it. Describe a candidate thunderstorm-related mechanism.**

Station table:
Ahead of boundary | Original wind, lower pressure
Behind boundary | Shifted wind, rising pressure

Answer: _____
_____
_____

**19. (2 points) Radar detects strong winds perpendicular to its beam but a small radial velocity. Explain why the wind need not be weak.**

Radar geometry table:
Wind direction | Perpendicular to beam
Radial velocity | Small
Independent wind observation | Strong

Answer: _____
_____
_____

**20. (2 points) A short intense rain period falls on an already saturated basin. Explain why rainfall amount alone does not establish flood severity.**

Basin table:
Antecedent soil | Saturated
Rain | Short, intense burst
Catchment drainage/routing | Not supplied

Answer: _____
_____
_____

**21. (2 points) Use Figure F1. Read the velocity difference between the adjacent west and east cells.**

## Same small storm area: collocated measurements

| Region | Reflectivity | Radial velocity | Correlation |
|---|---|---|---|
| West cell | 65 dBZ | -30 m/s | 0.70 |
| East cell | 60 dBZ | +35 m/s | 0.72 |
| Background | 25 dBZ | +5 m/s | 0.98 |

F1. Synthetic dual-polarization radar panels (original synthetic teaching figure)

Answer: _____
_____
_____

**22. (2 points) Use Figure F1. Explain why the juxtaposed velocity signs support investigating rotation rather than concluding that both cells are calm.**

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

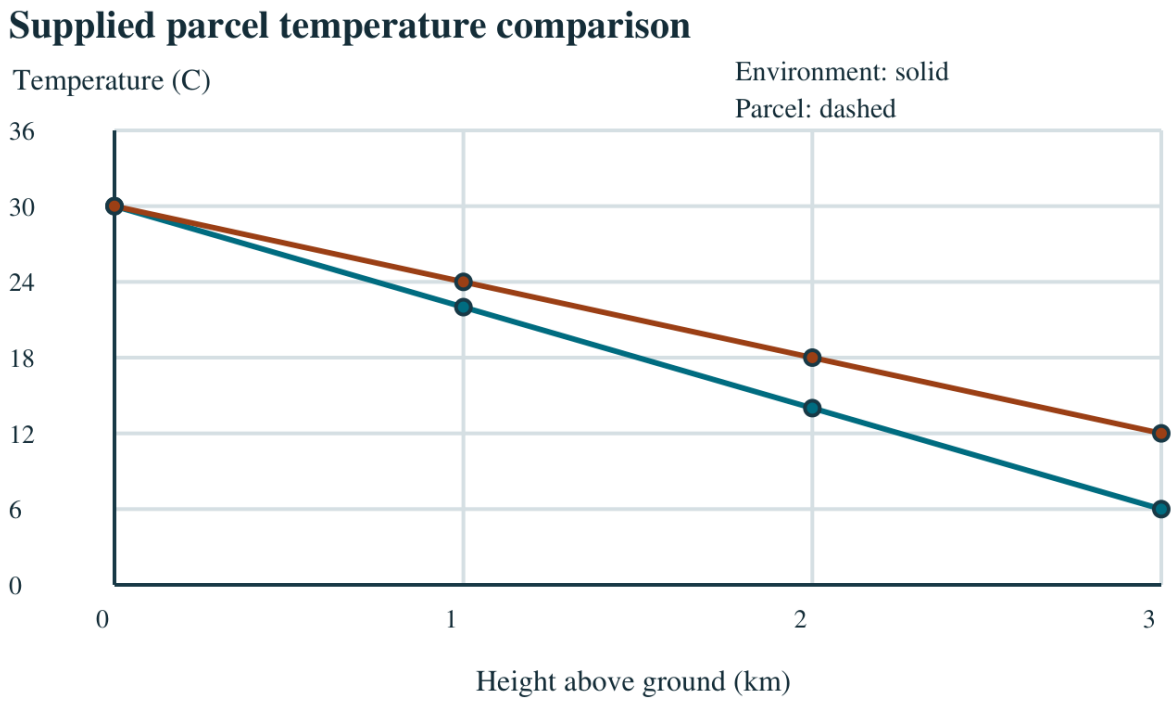
Answer: _____
_____
_____

**23. (2 points) Use Figure F1. What extra interpretation does low correlation in the same area support, and what does it not prove by itself?**

Refer to F1. Synthetic dual-polarization radar panels (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**24. (2 points) Use Figure F2. Read parcel-minus-environment temperature at 2 km.**




F2. Parcel and environment profiles (original synthetic teaching figure)

Answer: _____
_____
_____

**25. (2 points) Use Figure F2. Under the stated simplification, is the parcel positively or negatively buoyant at 2 km? Explain.**

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

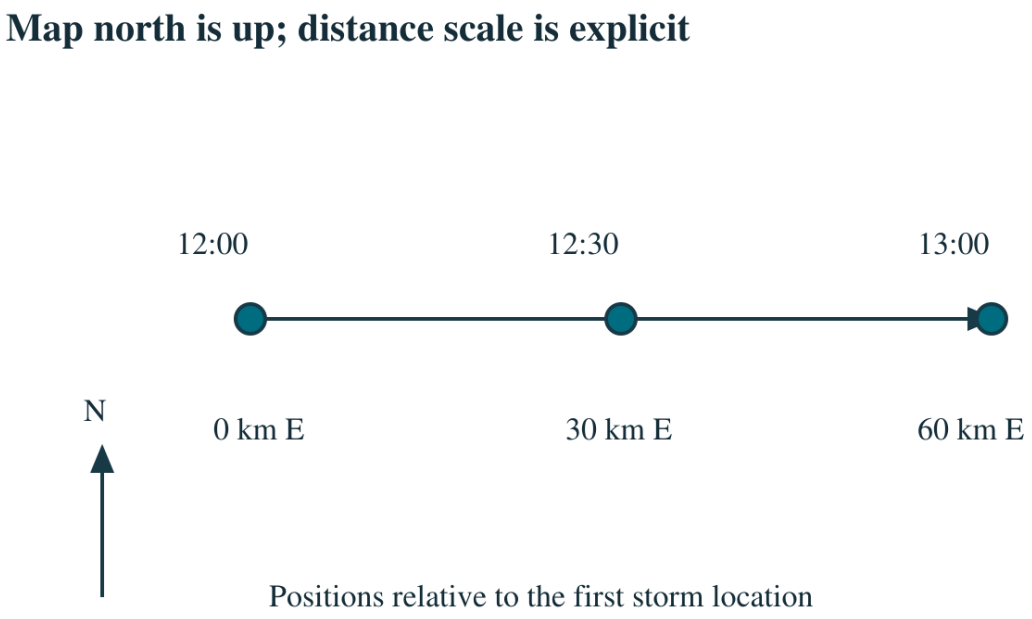
Answer: _____
_____
_____

**26. (2 points) Use Figure F2. Why cannot this profile alone establish tornado occurrence?**

Refer to F2. Parcel and environment profiles (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**27. (2 points) Use Figure F3. Calculate mean storm speed from 12:00 to 13:00.**




F3. Storm-track map (original synthetic teaching figure)

Answer: _____
_____
_____

**28. (2 points) Use Figure F3. Give the direction of the plotted motion.**

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

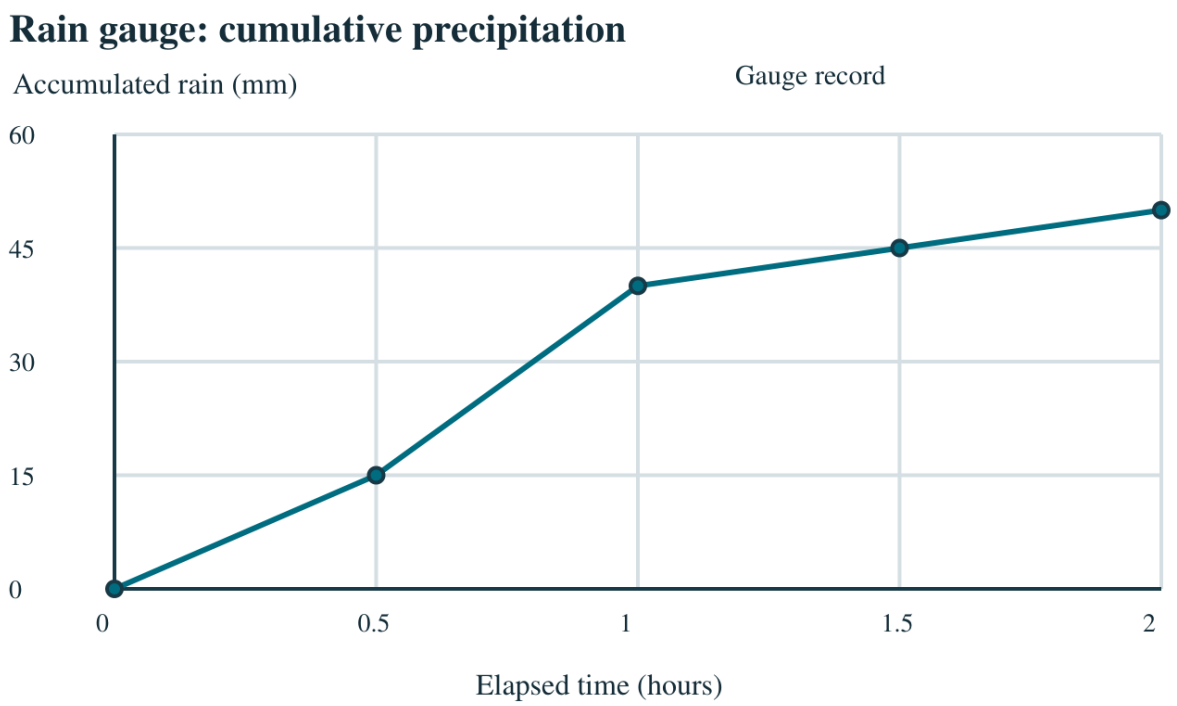
Answer: _____
_____
_____

**29. (2 points) Use Figure F3. If the motion continues unchanged, how many minutes after 13:00 would a point 30 km farther east be reached?**

Refer to F3. Storm-track map (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**30. (2 points) Use Figure F4. Calculate the mean rain rate during the 0.5-to-1.0 h interval.**




F4. Accumulated rainfall curve (original synthetic teaching figure)

Answer: _____
_____
_____

**31. (2 points) Use Figure F4. Identify the interval with the steepest accumulation and explain what the slope measures.**

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

**32. (2 points) Use Figure F4. For an already saturated basin, explain why the steep interval deserves attention even if a later interval has a higher cumulative total.**

Refer to F4. Accumulated rainfall curve (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____